

FORMAL EDUCATION

Louisiana State University Ph.D Student	Baton Rouge, Louisiana 2024–present
Universidad EAFIT M.S. in Biociences	Medellín, Colombia 2021–2023
– Thesis: “Identification of non-model mammal species using the MinION DNA sequencer from Oxford Nanopore.”	
Universidad EAFIT BS. in Biology	Medellín, Colombia 2016–2020
– Thesis: “Niche conservatism through evolutionary Time: A hierarchical analysis of niche divergence in a clade of Andean birds (Cotingidae)”	

WORK/TEACHING EXPERIENCE

Museum of Natural Science, Louisiana State University Curatorial Assistant - Ornithology Collection	Baton Rouge, Louisiana January 2026 - Present
Department of Biological Sciences, Louisiana State University Teaching Assistant	Baton Rouge, Louisiana August 2024 - December 2025
– Introduction to Biology (BIOL 1208) laboratory	
– Ecology (BIOL 4254) laboratory	
School of Applied Sciences and Engineering, Universidad EAFIT Lecturer	Medellín, Colombia January 2023 - July 2024
– Fundamentals of Computational Biology Course	
Department of Innovation and Technological Development, U. EAFIT Graduate research assistant	Medellín, Colombia August 2023 - July 2024
– Standardized laboratory methodologies for massive and cost-effective DNA sequencing, using the MinION from Oxford Nanopore Technologies	
Corporación Mustela Biotic professional	Olaya, Antioquia, Colombia May 2023
– Designed and structured the registration document for Socolao Civil Society Nature Reserve.	
– Characterization of bird species present in the Reserve.	
– Identified bird species subject to conservation in the reserve.	
– Developed the environmental zoning and management plan for the biotic component of the Socolao Civil Society Nature Reserve.	
– Accompanied the process of registering the reserve with National Natural Parks.	
Department of Biological Sciences, Universidad EAFIT Graduate teaching assistant	Medellín, Colombia August 2021 - 2023

- Directed the development of academic and laboratory activities of the campus Birdwatching group.

Department of Biological Sciences, Universidad EAFIT

Medellín, Colombia

Ornithology professional

May 2021 - July 2021

- Analyzed birds diversity data for the project “Biodiversity inventories from Avofruit SAS avocado crops” in the Colombian departments of Caldas and Risaralda.
- Wrote research article derived from the investigation of avifauna associated with forest remnants immersed in avocado plantations.

Terrasos SAS

Medellín, Colombia

Biotic professional

October 2020 - May 2021

- Characterized bird diversity in “El Silencio” and “La Laguna” farms (Angelópolis, Antioquia, Colombia).
- Characterized bird diversity in “Palo Hueco” farm (Liborina, Antioquia, Colombia).
- Wrote the official registration document of the Habitat Bank “El Globo” before the Colombian National Authority of Environmental Licenses (ANLA).

O2 Reserve

Santa Marta, Colombia

Biotic professional

September 2021 - November 2021

- Designed and structured the registration document for the Las Nubes Civil Society Nature Reserve.
- Characterization of bird species present in the Reserve.
- Identified bird species subject to conservation in the reserve.
- Developed the environmental zoning and management plan for the biotic component of the Las Nubes Civil Society Nature Reserve.
- Accompanied the process of registering the reserve with National Natural Parks.

Department of Biological Sciences, Universidad EAFIT

Medellín, Colombia

Undergraduate teaching assistant

October 2020 - May 2021

- Supported the development of academic and laboratory activities in the subject area Vertebrate zoology and systematics.

RESEARCH EXPERIENCE

Department of Biological Sciences, Universidad EAFIT

Medellín, Colombia

Graduate research assistant

August 2021 - Present

- Working in research project “Onsite and real-time cryptic species identification using the MinION DNA sequencer and its application in a conservation context.”
- Advisor: Juan F. Díaz-Nieto

Department of Biological Sciences, Universidad EAFIT

Medellín, Colombia

Undergraduate research assistant

2018 - 2019

- Developed research project “Sustainable development and avitourism in the Colombian Chocó”
- Advisors: Juan F. Díaz-Nieto and Alejandro Tobón-Valencia
- webpage: <https://erratico.info/Mecana>

Department of Biological Sciences, Universidad EAFIT

Antioquia, Colombia

Undergraduate ornithology research assistant

August 2018 - May 2019

- Participated in research project “BIO Anorí Expedition, Ministry of Science, Technology and Innovation (Government of Colombia).”
- Advisor: Paulo Pulgarín Restrepo

Department of Biological Sciences, Universidad EAFIT

Undergraduate mammalogy research assistant

Antioquia, Colombia

January 2017 - July 2017

- Participated in research project “Characterization of small mammals diversity in the Natural National Park Las Orquídeas, Colombia”
- Advisor: Juan F. Díaz-Nieto

PUBLICATIONS

- [1] **S. Velásquez-Restrepo**, M. C. Orozco, N. D. Franco-Sierra, J. M. Martínez-Cerón, and J. F. Díaz-Nieto, “Identification of non-model mammal species using the minion dna sequencer from oxford nanopore”, *PeerJ*, vol. 12, e17887, 9 Sep. 2024, ISSN: 21678359.
- [2] M. F. Velde, E. M. Besozzi, B. A. Krochuk, K. M. Henderson, B. R. Tsuru, **S. Velásquez-Restrepo**, H. M. Garrod, and J. C. Cooper, “What constitutes a community? a co-occurrence exploration of the costa rican avifauna”, *Neotropical Biodiversity*, vol. 9, pp. 64–75, 1 Dec. 2023, ISSN: 2376-6808.
- [3] **S. Velásquez-Restrepo**, M. Giraldo-Amaya, S. Saavedra-Porras, and J. F. Díaz-Nieto, “Avifauna asociada a bosques nativos inmersos en agroecosistemas de aguacate hass en caldas y risaralda (colombia)”, *Biota Colombiana*, vol. 24, e1070, 1 Dec. 2022, ISSN: 0124-5376.
- [4] P. Pulgarín-R, M. Olivera-Angel, L. Ortiz, D. Nanclares, **S. Velásquez-Restrepo**, and J. Díaz-Nieto, “Dna barcodes of birds from northern colombia”, *Biodiversity Data Journal*, vol. 9, e64842, May 2021, ISSN: 1314-2828.

SCHOLARSHIPS AND AWARDS

- Virginia Mouw Award, LSU Museum of Natural Science September 2025–Present
- Fulbrighth Scholarship, Ministerio de Ciencia Tecnología e Innovación (Gobierno de Colombia) and Fulbright Commission Colombia Cohort 2024
- Colombia Biodiversa Grant, Fundación Alejandro Angel Escobar January 2023–February 2025
- Research assistantship, Universidad EAFIT January 2022–July 2024
- Research and creation grant, Universidad EAFIT February 2019 –January 2020

PROFESSIONAL MEMBERSHIPS

- American Ornithological Society February 2025–Present
USA
- Society for the Study of Evolution February 2025–Present
USA
- Niche Modelling research group May 2020–December 2021
Chicago, USA
- Research group “Biodiversidad, Evolución y Conservación (BEC)” July 2019–Present
Department of Biological Sciences, Universidad EAFIT. Medellín Colombia
- Asociación Colombiana de Zoología December 2018–Present
Bogotá, Colombia

LANGUAGES

- **Spanish:** Native
- **English:** TOEFL iBT 113 points (Reading: 29, Listening: 28, Speaking: 26, Writing: 30).
- **French:** DELF B2

AREAS OF INTEREST

- Evolutionary biology
- Molecular phylogenetics
- Phylogeography
- Molecular biology
- Avian and mammalian systematics

COMPUTATIONAL SKILLS

- **R:** Good
- **Bash:** Good
- **Qgis:** Good
- **Excel:** Good
- **Geneious:** Good
- **Photoshop:** Basic
- **Lightroom:** Basic

OTHER RELEVANT SKILLS

- **Molecular Biology:**
 - DNA extraction
 - DNA amplification (Conventional PCR)
 - DNA sequencing - Oxford Nanopore Technologies
- **Biological collection specimens preparation (birds and small mammals).**

CONGRESS AND WORKSHOPS

VIII Colombian Ornithology Congress

December 2025

Universidad del Valle. Cali, Colombia

- Poster Session: “Carotenoid gene evolution and hybrid zone dynamics in *Manacus manakins* of northwestern Colombia”.

VI Colombian Zoology Congress

March 2023

Universidad de Córdoba. Montería, Colombia

- I Symposium on Molecular Taxonomy, Beyond Traditional Taxonomy: “Códigos de barras de ADN y su buen uso en la taxonomía integrativa del siglo XXI”
- I Symposium on Molecular Taxonomy, Beyond Traditional Taxonomy: “Hacia la identificación in situ y en tiempo real de especies crípticas de pequeños mamíferos utilizando el secuenciador de ADN MinION.”

Niche Modeling Workshop

April-May, 2019

Universidad EAFIT-Universidad Nacional Autónoma de México(UNAM) - Universidad CES - Universidad de Antioquia. Medellín, Colombia

IV Colombian Zoology Congress

December 2018

Instituto de Investigaciones Biológicas Alexander Von Humboldt. Bogotá, Colombia

DNA Barcodes Workshop

June, 2018

Universidad EAFIT. Medellín, Colombia